

Finding appropriate bias correction methods in downscaling precipitation for hydrologic impact studies over North America

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[1] This work compares the performance of six bias correction methods for hydrological modeling over 10 North American river basins. Four regional climate model (RCM) simulations driven by reanalysis data taken from the North American Regional Climate Change Assessment Program intercomparison project are used to evaluate the sensitivity of bias correction methods to climate models. The hydrological impacts of bias correction methods are assessed through the comparison of streamflows simulated by a lumped empirical hydrology model (HSAMI) using raw RCM-simulated and bias-corrected precipitation time series. The results show that RCMs are biased in the simulation of precipitation, which results in biased simulated streamflows. All six bias correction methods are capable of improving the RCM-simulated precipitation in the representation of watershed streamflows to a certain degree. However, the performance of hydrological modeling depends on the choice of a bias correction method and the location of a watershed. Moreover, distribution-based methods are consistently better than mean-based methods. A low coherence between the temporal sequences of observed and RCM-simulated (driven by reanalysis data) precipitation was observed over 5 of the 10 watersheds studied. All bias corrections methods fail over these basins due to their inability to specifically correct the temporal structure of daily precipitation occurrence, which is critical for hydrology modeling. In this study, this failure occurred on basins that were distant from the RCM model boundaries and where topography exerted little control over precipitation. These results indicate that bias correction performance is location dependent and that a careful validation should always be performed, especially on studies over new regions.

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1. Introduction

[2] The impacts of climate change on watershed hydrology are of great interest to water resource managers and hydropower shareholders [Arnell *et al.*, 2001; Bates *et al.*, 2008]. General Circulation Models (GCMs) are the major tools that project future climate change information. However, GCMs' outputs cannot be directly used to force hydrological models for assessing the hydrological impacts of climate change [Sharma *et al.*, 2007; Hansen *et al.*, 2006; Feddersen and Andersen, 2005; Chen *et al.*, 2011a], because GCMs do not provide reliable information on scales below ~200 km for most hydrological-relevant variables [Maraun *et al.*, 2010]. In particular, in mountainous regions, fine-scale information needs to be incorporated

into coarse GCM projections for hydrologic modeling because of the altitude-dependence of precipitation and temperature that are usually not represented by GCMs [Seager and Vecchi, 2010].

[3] Dynamical downscaling is one of the techniques that transfer information from GCMs to finer scales by applying a higher-resolution regional climate model (RCM) over a limited area with initial and boundary conditions taken from a driving GCM [Dickinson *et al.*, 1989; Giorgi, 1990; Yang *et al.*, 2010]. Compared to GCMs, RCMs are more reliable at reproducing relevant mesoscale patterns of local precipitation, and so the topographic effects on precipitation can be much better represented at a regional scale [Frei *et al.*, 2003, 2006; Intergovernmental Panel on Climate Change, 2007; Buonomo *et al.*, 2007]. However, RCMs are still subjected to considerable biases when comparing the simulated climate at the reference period with the same period's observations, even at the same scale. For example, most RCMs tend to overestimate the frequency of precipitation and the occurrence of light precipitation, while they underestimate the heavy precipitation [Murphy, 1999; Fowler *et al.*, 2007]. Furthermore, the accuracy of an RCM is seasonally and regionally dependent [Kotlarski *et al.*, 2005; Maraun *et al.*, 2010]. Generally speaking, RCMs show diminished accuracy at representing summer

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precipitation than autumn and winter precipitation, due to the difficulties in modeling convective rainfall [Maraun *et al.*, 2010]. RCMs perform better in regions with temperate climate conditions than tropical regions, since tropical rainfall is often predominantly convective in nature and occurs at a subdaily time scale [Lenderink and van Meijgaard, 2008]. RCM biases can be introduced by incorrect boundary conditions provided by reanalysis data or by a GCM, and by systematic model errors, such as the errors resulting from the imperfect parameterization of some climate processes [Ehret *et al.*, 2012; Teutschbein and Seibert, 2012].

[4] Therefore, postprocessing of RCM outputs (such as bias correction) is normally a prerequisite step of most climate change impact studies. Bias correction is a model output statistics approach that seeks to use information from biased model outputs. Several bias correction methods ranging from simple scaling to sophisticated distribution mapping have been developed in the last decade [Sharma *et al.*, 2007; Piani *et al.*, 2010; Mpelasoka and Chiew, 2009; Ryu *et al.*, 2009; Chen *et al.*, 2011b, 2013; Salvi *et al.*, 2011; Izumi *et al.*, 2011; Lafon *et al.*, 2012; Teutschbein and Seibert, 2012]. It is necessary to evaluate and compare their performances for a specific impact study.

[5] Piani *et al.* [2010] validated a bias correction method (distribution mapping based on the gamma distribution) for correcting RCM-simulated daily precipitation over Europe. The results showed that this method performed reasonably well, not only at the mean but also at other moments (drought index and heavy precipitation) of intensity distribution. Terink *et al.* [2009] corrected RCM-simulated precipitation by fitting the mean and coefficient of variation of the observation, and corrected RCM-simulated temperature by fitting the mean and standard deviation of the observation. This bias correction process led to satisfactory results as precipitation and temperature differences between RCM data and observations decreased significantly. Bennett *et al.* [2011] tested the performance of a quantile mapping bias correction method (based on an empirical distribution) for use in hydroclimatological projections in Australia. The bias correction substantially improved the spatial correlation between modeled and observed seasonal and annual rainfall. This method can effectively couple RCM outputs to a hydrological model for assessing the hydrological impact in a changing climate. Marke *et al.* [2011] evaluated the effect of two bias correction methods on hydrological modeling. These two bias correction approaches allowed for a more accurate simulation of discharge, and the correction for subgrid variability was found to reduce biases in simulated discharge compared to RCM data without bias correction. Overall, bias correction methods are generally capable of reducing errors in RCM outputs [Terink *et al.*, 2009; Bennett *et al.*, 2011; Themeßl *et al.*, 2010]. Further benefits were also found in hydrological impact studies. However, bias correction methods may perform differently in projecting climate change scenarios and quantifying climate change impacts [Themeßl *et al.*, 2010; Teutschbein and Seibert, 2012].

[6] Lafon *et al.* [2012] compared the performance of four bias correction methods (linear, nonlinear, gamma, and empirical distribution-based quantile mapping) in the reduction of biases in RCM-simulated precipitation for

seven catchments spread across Great Britain. The results showed that the mean and standard deviation of daily precipitation can be corrected robustly; while the correction of skewness and kurtosis of daily precipitation are much more sensitive to the choice of a bias correction method and the selection of a particular calibration period. If both precipitation data sets (modeled and observed) can be approximated by a gamma distribution, the gamma-based quantile mapping method offers the best combination of accuracy and robustness. Otherwise, the nonlinear method is more effective at reducing the bias. Gudmundsson *et al.* [2012] compared the performance of distribution-derived transformations, parametric transformations, and nonparametric transformations at downscaling precipitation for 83 stations in Norway. Nonparametric transformations gave the best performance in the reduction of systematic biases in RCM-simulated precipitation. The above mentioned comparisons were only conducted on the construction of climate projection, which is the first step for impact studies.

[7] Other studies included the comparison of different bias correction methods in the assessment of climate change impacts. Mpelasoka and Chiew [2009] compared the impact of three empirical downscaling methods, constant scaling (also known as the delta change method), daily scaling, and daily translation (DT), on the construction of runoff projections across Australia. The results showed that the differences between constant and daily scaling are very small in constructing the mean annual runoff. However, daily scaling and DT methods are generally better at representing the extreme runoff than constant scaling, because they both specifically take into account the increase in extreme daily rainfall. Themeßl *et al.* [2010] compared an ensemble of seven statistical and bias correction approaches in downscaling RCM daily precipitation over the historical period in the Alps region. The results showed that bias correction approaches such as quantile mapping and local intensity (LOCI) scaling displayed significant advantages compared to the traditional multiple linear regression methods. The quantile mapping method showed the best performance over the other methods, particularly in the downscaling of precipitation extremes. However, van Roosmalen *et al.* [2011] showed that the delta change and distribution mapping methods were not significantly different in the projection of changes in mean hydrological responses for a Denmark catchment. However, that study only analyzed the mean hydrological statistics; the response of extreme hydrological events may be different depending on the choice of bias correction method. Chen *et al.* [2013] assessed the uncertainty of six empirical downscaling methods (two change factor approaches and four bias correction approaches) in quantifying the hydrological impact of climate change over two North American river basins. They concluded that the empirical downscaling method contributes great uncertainty to the projection of statistics of extreme events for both climate change projection and its simulated streamflow, and that the uncertainty linked to the choice of a change factor or bias correction approach is larger than within each type. However, this study was incomplete because it only assessed the uncertainty of empirical downscaling methods for future climate change impacts over two North American watersheds. The performance of different bias

correction approaches was not compared under the recent past climate conditions. *Teutschbein and Seibert* [2012] evaluated four bias correction methods for precipitation and three bias correction methods for temperature in assessing the hydrological impacts of climate change for five catchments in Sweden, with areas ranging between 147 and 293 km². The results showed that all the bias correction methods were able to improve the RCM outputs to some extent. However, the corrected precipitation and temperature performance depended on the choice of a correction method. Distribution mapping was found to be the best correction method for both climate projections and hydrological impact qualifications; it performed especially well in terms of the simulation of hydrological extremes. This is probably one of the most thorough studies to date in terms of inclusion of the most bias correction approaches for hydrological impact studies. However, this study was only conducted over Sweden for five small catchments. Even though the catchments were selected from different climatic conditions, most of Sweden has a temperate climate. Furthermore, the performance of bias correction methods may be different between small-scale and meso-scale or large-scale catchments, due to a spatial average of climate model outputs.

[8] North America is the world's third-largest continent and includes various climate conditions from tropical to polar. The reliability of RCM outputs and performance of bias correction may be regionally dependent. Thus, it is of great importance to know the performance of bias correction methods and their regional variability for quantifying hydrologic impacts over North America. This work evaluates the performance of six bias correction methods with respect to hydrological modeling over North America. The six bias correction methods can be classified into mean-based (two methods) and distribution-based (four methods) approaches. Alternatively, they can also be classified into approaches with (four methods) and without (two methods) correction in wet-day frequency of precipitation. Those six methods cover most of the existing bias correction methods. In addition, compared to other studies using RCM outputs driven by GCMs, such as *Teutschbein and Seibert* [2012], this work uses four RCM simulations driven by reanalysis data over the recent past. Thus, this makes it possible to assess the hydrological responses of RCM outputs with or without bias correction at the daily scale. The studies that used RCM outputs driven by GCMs cannot uncover the effect of the temporal structure of daily precipitation occurrence on the performance of a bias correction method. Precipitation, a critical element of hydrology studies that is much more difficult to be simulated by RCMs than temperature, is the only variable evaluated in this study. Moreover, the correction of precipitation rather than temperature (using the "perfectly" observed temperature for hydrological modeling) makes it possible to evaluate the performance of bias correction methods on only precipitation and avoid the influence from the corrected temperature. Ten river basins dispersed across several climate zones and topographic conditions over Canada and the United States (US) were selected to represent the natural climate variability in North America and to enable an assessment of the spatial variability of the performance of bias correction methods.

2. Study Area and Data Set

2.1. Study Area

[9] Ten river basins dispersed across Canada and the US were selected for this study (Figure 1). All 10 river basins are located within the North American domain of the RCM, with three basins located in Canada and seven located in the US. The 10 basins cover five climate zones based on the Köppen climate classification [*Kottek et al.*, 2006] for North America. The 10 basins were also selected from different topographic conditions from the westerly mountainous area to the centrally and easterly flatter terrain. Table 1 presents the basic information of all 10 basins. The basin's areas are between 1271 and 24,610 km² representing meso-scale and large-scale catchments over North America. The yearly average precipitation and the maximum and minimum temperatures (Tmax and Tmin) range from 409 to 1428 mm, 2.5 to 24.7°C, and -7.9 to 11.1°C, respectively, which adequately represents climate variability over North America. The three basins in Canada are, respectively, located in the provinces of Quebec, Saskatchewan, and British Columbia, each with different climate conditions even though they are all snow-dominated basins. Carrot is the driest basin with mean annual precipitation of only 409 mm, and Manicouagan 5 (Manic 5) is the wettest and coldest basin with mean annual precipitation, Tmax and Tmin of 902 mm, 2.5°C, and -7.9°C, respectively. The seven US basins can be geographically divided into two groups: Three basins (Umpqua, Yellowstone, and Yampa) are located in westerly mountainous areas and the other four (Grand, Wolf, Chickasawhay, and Sacandaga) are located in the central, eastern, and southern US with much flatter terrain. These seven basins can also be classified, according to climate conditions, into four snow-dominated basins (Yellowstone, Yampa, Wolf, and Sacandaga) and three no-snowfall basins (Umpqua, Grand, and Chickasawhay). Umpqua is located on the west coast and Grand and Chickasawhay are located in the Central and Southern US, respectively.

2.2. Data Set

[10] Observed daily precipitation, Tmax and Tmin, and RCM-simulated daily precipitation are used in this study. The observed meteorological data for the three Canadian basins were taken from the National Land and Water Information Service (www.agr.gc.ca/nlwis-snite) data set. This data set was created by interpolating station data to a 10 km grid using a thin plate smoothing spline surface fitting method [*Hutchinson et al.*, 2009]. Discharge data for Manic 5 were obtained from mass balance calculations at the dam and were provided by Hydro-Quebec. The discharge of the other two Canadian basins (Nation and Carrot) was obtained from Environment Canada (<http://www.ec.gc.ca/>). The observed data (daily precipitation, Tmax, Tmin and discharge) for the US basins were obtained from the Model Parameter Estimation Experiment (MOPEX) database (http://www.nws.noaa.gov/oh/mopex/mo_datasets.htm) [*Duan et al.*, 2006]. The RCM-simulated precipitation was derived from the North American Regional Climate Change Assessment Program (NARCCAP) (<http://www.narccap.ucar.edu/>), an international program to produce high resolution climate change simulations to investigate uncertainties in regional scale projections of future climate and generate climate change scenarios for use in climate impact research [*Mearns et al.*, 2007, 2009]. Four RCMs

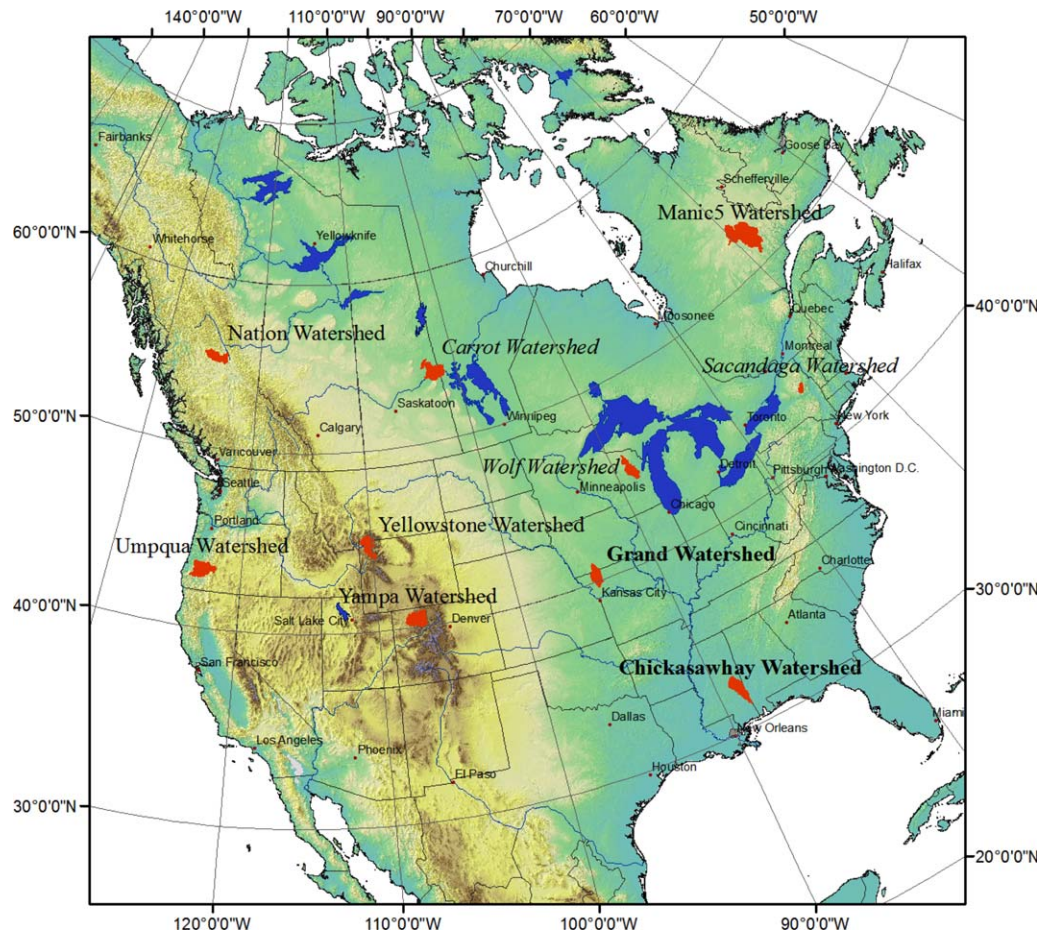


Figure 1. Location of the 10 selected river basins over North America. Different typenames on watershed name represent different performances of bias correction approaches for hydrological modeling: roman = good, italic = bad, and bold = very bad.

(Canadian Regional Climate Model (CRCM), Hadley Regional Model 3 (HRM3), Regional Climate Model version 3 (RCM3), and Weather Research & Forecasting model (WRF)) driven by the National Center for Environmental Prediction (NCEP) reanalysis data over the North American domain (Figure 1) were used in this study. All the RCMs were run at a spatial resolution of 50 km over the 1981–2000 period. Because the area of each basin is larger than one RCM grid box, basin-averaged precipitations were used. In fact, averaging over a few grid points is necessary even for

very small watersheds (such as used in the study of *Teutschbein and Seibert* [2012]) to eliminate the risk of grid-point numerical effect in the climate model computational schemes.

3. Methodology

3.1. Bias Correction Methods

[11] Six bias correction methods, linear scaling (LS), LOCI scaling, DT, daily bias correction (DBC), quantile mapping based on an empirical distribution (QME) and

Table 1. Basic Information of the Selected 10 River Basins Over North America and the Nash-Sutcliffe Coefficient (NSE) of the Calibration and Validation of Hydrological Model HSAMI^a

Country	Province or State	Basin Name	Area (km ²)	Precip (mm)	Tmax (°C)	Tmin (°C)	Calibration Period	NSE Calibration	Validation Period	NSE Validation
Canada	BC	Nation	6790	612	5.9	−4.3	1981–1993	0.932	1994–2003	0.905
	SK	Carrot	12,600	409	6.1	−5.4	1967–1980	0.822	1981–2000	0.719
	QC	Manic 5	24,610	902	2.5	−7.9	1966–1980	0.917	1981–2000	0.856
	OR	Umpqua	9535	1270	16.7	4.4	1948–1980	0.907	1981–2000	0.889
USA	MT	Yellowstone	6791	688	8.5	−5.3	1948–1980	0.897	1981–2000	0.701
	CO	Yampa	8828	598	12.5	−3.4	1948–1980	0.815	1981–2000	0.728
	MO	Grand	5825	872	16.9	4.9	1948–1980	0.754	1981–2000	0.778
	WI	Wolf	5851	769	12.0	0.7	1948–1980	0.847	1981–2000	0.744
	MS	Chickasawhay	6967	1428	24.7	11.1	1948–1980	0.861	1981–2000	0.851
	NY	Sacandaga	1271	1215	11.7	−0.5	1948–1980	0.798	1981–2000	0.807

^aPrecip = precipitation, Tmax = maximum temperature, and Tmin = minimum temperature.

quantile mapping based on a gamma distribution (QMG) are compared in this study. The six methods can be classified into mean-based approaches (LS and LOCI) and distribution-based approaches (DT, DBC, QME, and QMG). Alternatively, they can also be classified into approaches with (LOCI, DBC, QME, and QMG) and without (LS and DT) correction in wet-day frequency. All bias corrections were conducted on a monthly basis. The observed precipitation threshold for the determination of a wet day was 0.5 mm. Four of the six bias correction methods used in this study were also used by *Chen et al.* [2013] to assess the uncertainty of empirical downscaling approaches in quantifying the hydrological impact of future climate change over two North American watersheds. Table 2 briefly describes the characteristics of all six bias correction methods. More details can be found in *Chen et al.* [2013] and from the references listed in Table 2.

3.2. RCM Without Bias Correction

[12] In this method, no bias correction is made, but the hydrology model is specifically calibrated to RCM data against observed discharge (referred to as specific calibration). The methodology and theory behind this method can be found in *Chen et al.* [2011b]. The assumption behind this method is that the biases of RCM outputs are sufficiently small so that they can be overcome by the hydrology model through the calibration process. It is well known that observed precipitation and temperature data can also be biased. If the calibrated hydrological model with RCM outputs is able to adequately reproduce the observed discharge with realistic internal parameters, it can be argued that RCM data is no more biased than observed data. Thus, they can be used directly for hydrological impact studies. As this work only evaluates the performance of precipitation bias correction, RCM-simulated precipitation combined with observed Tmax and Tmin are used to calibrate the hydrological model.

3.3. Hydrological Simulation

[13] Flow discharge was simulated using the hydrological model HSAMI, developed by Hydro-Québec. HSAMI has been used to forecast natural inflows for over 20 years [Fortin, 2000]. It is used by Hydro-Québec for hourly and daily forecasting of natural inflows on nearly 100 watersheds with surface areas ranging from 160 to 69,195 km². HSAMI is a 23 parameter, lumped, conceptual, rainfall-runoff model. Two parameters account for evapotranspiration, six for snowmelt, 10 for vertical water movement, and five for horizontal water movement. Vertical flows are simulated with four interconnected linear reservoirs (snow on the ground, surface water, unsaturated, and saturated zones). Horizontal flows are routed through two-unit hydrographs and one linear reservoir. Model calibration was done automatically using the Covariance Matrix Adaptation Evolution Strategy [Hansen and Ostermeier, 1996, 2001]. The model takes snow accumulation, snowmelt, soil freezing/thawing, and evapotranspiration into account.

[14] The basin-averaged minimum required daily input data for HSAMI are: Tmax, Tmin, liquid, and solid precipitations. Cloud cover fraction and snow water equivalent can also be used as inputs, if available. A natural inflow or

discharge time series is also needed for proper calibration/validation.

[15] The time periods of observed precipitation and temperature used to calibrate (referred to as standard calibration) and validate the HSAMI are presented in Table 1 for all 10 river basins. The validation covers the 1981–2000 period, with the exception of the Nation basin, where the validation period covers the period 1994–2003 (due to data availability). The optimal combination of parameters was chosen based on the Nash-Sutcliffe coefficient (NSE). The mean annual hydrograph of the observed (Qobs) and simulated (Qsim) hydrographs show the small biases introduced by the hydrological model (Figure 2). The overall fit is very good, with NSE values ranging between 0.754 and 0.932 for calibration and between 0.701 and 0.905 for validation across all 10 basins (Table 1). The high NSE values are representative of the good quality of weather inputs and observed discharge for all basins. The mean annual hydrographs also show that the 10 basins can be classified into two groups: Snow-dominated basins (Nation, Carrot, Manic 5, Yellowstone, Yampa, Wolf, and Sacandaga) and no-snowfall basins (Umpqua, Grand, and Chickasawhay). The hydrological regime can be divided into dry and wet seasons for Umpqua and Chickasawhay.

3.4. Results Analysis

[16] The evaluation of the bias correction methods is based on a split-sample cross-validation approach [Bennett et al., 2011] with the following four steps. (1) The 20 years (1981–2000) of observed and RCM-simulated precipitation was first separated into two periods: 10 odd years and 10 even years. (2) The 10 odd years were used to calibrate the bias correction methods and the 10 even years used as a validation period. (3) In an opposite sense, the 10 even years were used to calibrate the methods and the 10 odd years were used as the other validation period. (4) Two 10 year validation periods (10 even years from step (2) and 10 odd years from step (3)) were combined continuously to construct a 20 year validation period. The split-sample cross-validation approach is commonly used in hydrology studies. It is used to reduce the risk of having a model overfitted to a period affected by the interannual variability of the climate system. The goal is the same for the downscaling method. This study aims at answering the following questions: (1) Are the complicated distribution mapping and the correction in wet-day frequency necessary for accurate precipitation reproduction and hydrological modeling? (2) Is the performance of bias correction methods watershed-dependent? (3) What is the best correction method for hydrological studies?

[17] The constructed 20 years of precipitation values with bias correction were then compared with the observed precipitation using mean, standard deviation, and 95th percentiles (heavy precipitation) of daily precipitation as criteria. The RCM-simulated precipitation without bias correction was also compared with observed data and bias corrected data.

[18] Following bias correction, the 20 year time series of precipitation were combined with the observed Tmax and Tmin. These series are then used to drive HSAMI to simulate 20 years of daily discharge time series. The raw RCM-simulated precipitation combined with the observed Tmax

Table 2. The Advantages (or Characteristics) and Disadvantages of the Six Bias Correction Methods

Method	Advantage	Disadvantage	Reference
1. Linear scaling (LS)	<i>Mean-based</i> A mean monthly correction factor is applied to the RCM-simulated daily precipitation in a month. It is the simplest bias correction method.	The daily precipitation sequence is the same as that of the RCM-simulated data (usually too many wet days compared to the observation). It does not account for the changes in the frequency distribution of precipitation. No adjustment is made to the temporal structure of daily precipitation occurrence.	<i>Lenderink et al. [2007]</i> and <i>Teutschbein and Seibert [2012]</i>
2. Local intensity scaling (LOCI)	<i>Mean-based</i> The wet-day frequency is corrected. A mean monthly correction factor is applied to the RCM-simulated daily precipitation in a month.	It does not account for the different changes in the frequency distribution of precipitation. No adjustment is made to the temporal structure of daily precipitation occurrence.	<i>Schmidli et al. [2006]</i>
3. Daily translation (DT)	<i>Distribution-based</i> Different correction factors (differences in percentiles between observed and RCM-simulated data at the calibration period) are applied to correct the frequency distribution of RCM-simulated daily precipitation.	The daily precipitation sequence is the same as that of the RCM-simulated data (usually too many wet days compared to the observation). No adjustment is made to the temporal structure of daily precipitation occurrence.	<i>Mpelasoka and Chiew [2009]</i>
4. Daily bias correction (DBC)	<i>Distribution-based</i> Combines the DT and LOCI approaches to take into account the different changes in frequency distributions of daily precipitation, and corrects the wet-day frequency of precipitation.	No adjustment is made to the temporal structure of daily precipitation occurrence.	<i>Schmidli et al. [2006]</i> and <i>Mpelasoka and Chiew [2009]</i>
5. Quantile mapping based on an empirical distribution (QME)	<i>Distribution-based</i> Corrects the RCM-simulated precipitation based on point-wise daily constructed empirical cumulative distribution functions (ecdfs). The frequency of precipitation occurrence is corrected at the same time.	No adjustment is made to the temporal structure of daily precipitation occurrence.	<i>Thiemeßl et al. [2010, 2011]</i>
6. Quantile mapping based on a gamma distribution (QMG)	<i>Distribution-based</i> Corrects the RCM-simulated precipitation based on a gamma distribution. The frequency of precipitation occurrence is corrected using the LOCI method.	The performance depends on whether the observed and RCM-simulated precipitation follows the gamma distribution (or not). No adjustment is made to the temporal structure of daily precipitation occurrence.	<i>Ines and Hanson [2006]</i> , <i>Piani et al. [2010]</i> , and <i>Teutschbein and Seibert [2012]</i>

and T_{min} was also used to drive the hydrological model. Discharges simulated using bias-corrected precipitation were compared with those simulated using observed and raw RCM-simulated precipitation, using three hydrological statistics and three evaluation metrics. The hydrological statistics are the mean daily discharge, the mean of 95th percentile spring (April-May-June, AMJ) high flow, and

the mean of 5th percentile summer (July-August-September, JAS) low flow. The three evaluation metrics are the NSE, the Root Mean Square Error (RMSE), and the Transformed Root Mean Square Error (TRMSE) [van Werkhoven et al., 2009].

[19] Discharges simulated using raw RCM data with specific calibration of the hydrological model were also used

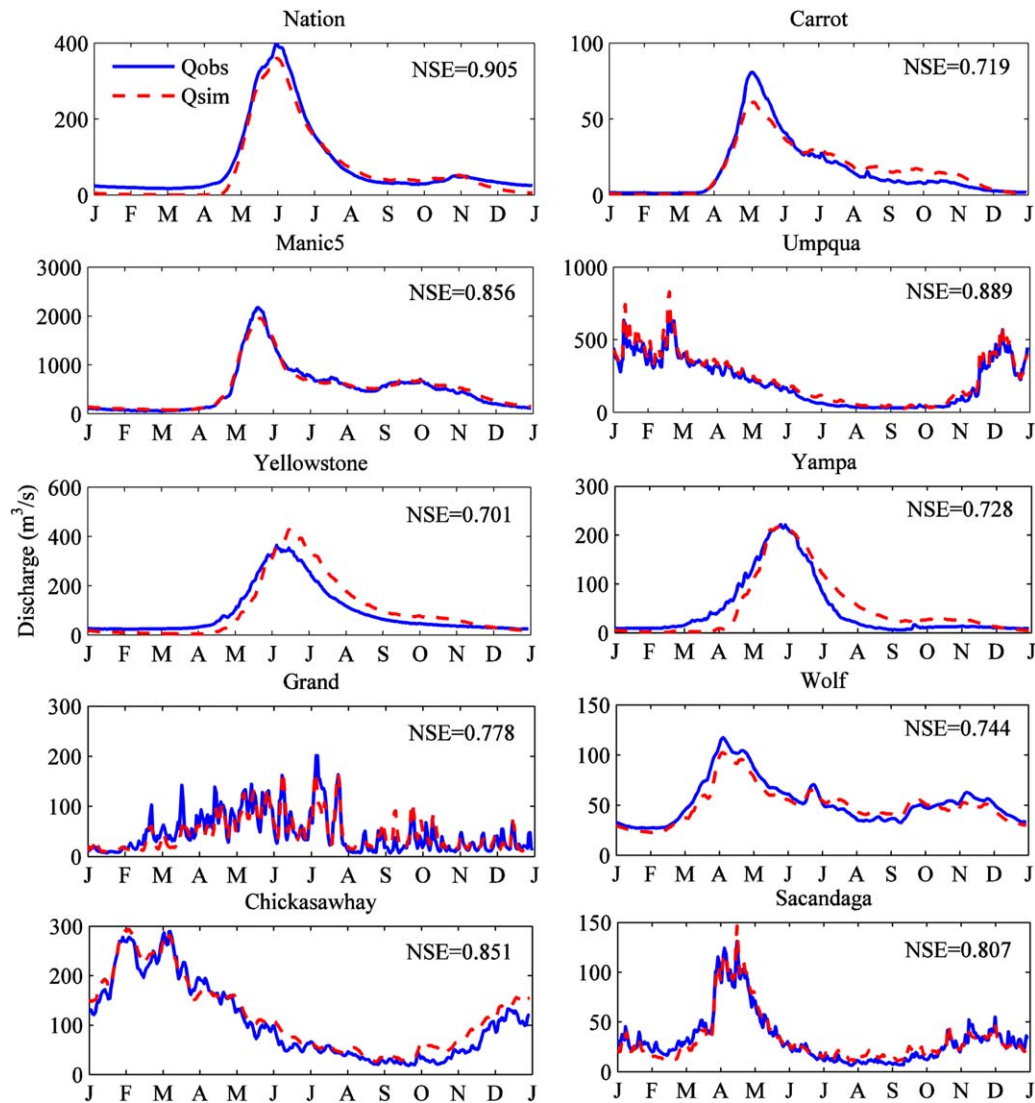


Figure 2. Mean annual hydrographs of observed discharge (Q_{obs}) and observed meteorological data (precipitation, and maximum and minimum temperatures) simulated discharge (Q_{sim}) for the validation period of 1981–2000. The Nash-Sutcliffe coefficients (NSE) over the 20 year validation period are presented in the figure (Note: the NS values are for the daily continuous data and not for the mean hydrographs).

for comparison. The discharge was also constructed using the same split-sample approach discussed above.

4. Results

4.1. Climate Simulation

[20] Figure 3 presents the absolute relative error (ARE) ($(P_{sim} - P_{obs}) / P_{obs} \times 100$) of mean wet-day precipitation simulated by four RCMs and corrected by six bias correction methods. Each boxplot is constructed using 10 mean values from 10 river basins. The results clearly show that, compared to the observed precipitation, the mean value of RCM-simulated precipitation is very biased, especially for HRM3 and WRF3. The LS and DT methods do not show improvement in the statistic of mean wet-day precipitation. This is to be expected because they do not take into account the overestimation of the RCM-simulated

wet-day frequency. Both methods correct the monthly mean precipitation, but the overestimation of wet days results in a smaller mean wet-day precipitation during the calculation. The other four bias correction methods do correct for wet-day frequency and this significantly improves the mean wet-day precipitation, with medians of AREs ranging between 0.7% and 3.4%, 1.7% and 2.6%, 1.2% and 4.4%, and 0.4% and 2.0% for the LOCI, DBC, QME, and QMG methods, respectively. Overall, even though the performance of RCMs is different in the simulation of mean wet-day precipitation (CRCM is better than the other models.), the four bias correction methods that correct for wet-day frequency successfully reduce the biases.

[21] None of the six bias correction methods specifically account for the correction in the variance of daily precipitation.

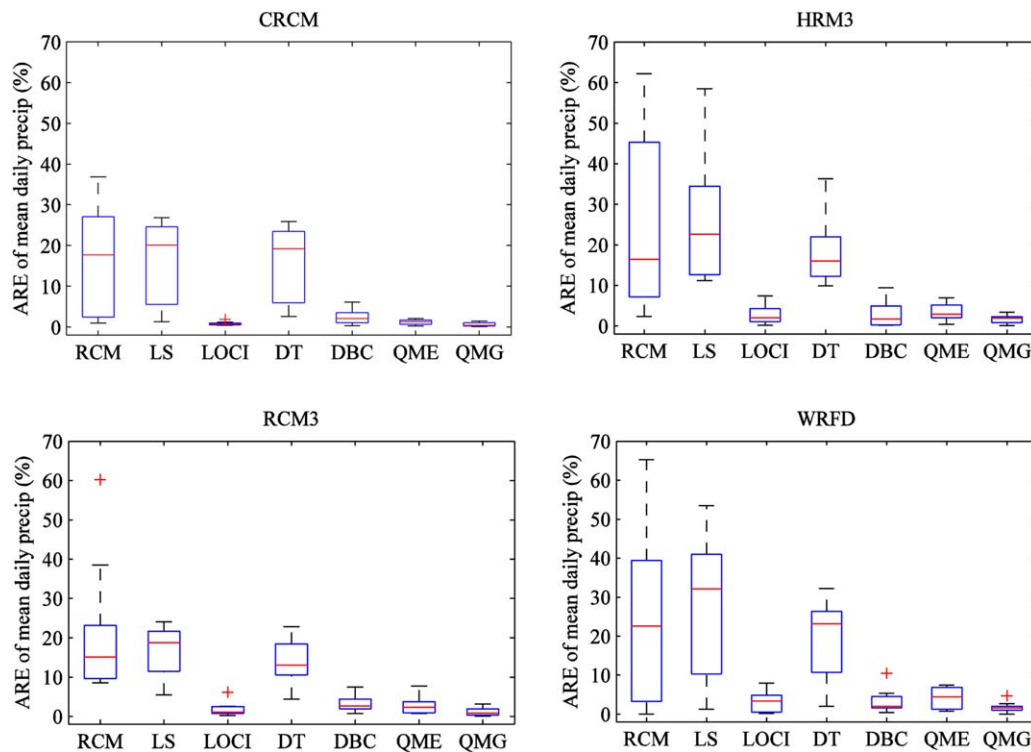


Figure 3. Boxplot of the absolute relative error (ARE) of raw RCM-simulated (RCM) and six bias correction method-corrected daily mean precipitation (precip) values for all four RCMs. Each boxplot is constructed using 10 mean values from 10 river basins.

However, the correction in mean precipitation does affect the variance. Figure 4 presents the AREs of standard deviation of the wet-day precipitation. Similarly to the mean wet-day precipitation, the standard deviation of RCM-simulated precipitation is biased. All the bias correction methods somewhat improve the standard deviation simulated by the CRCM. However, the standard deviation is not improved as much by the mean-based methods (LS and LOCI) for the other three RCMs. In particular, the LS method even introduced additional bias when using HRM3- and WRFG-simulated precipitation. However, the other four distribution-based methods are able to improve the standard deviation, even though the performances are different. QME and QMG perform consistently better than DT and DBC for HRM3, RCM3, and WRFG, with QMG performing the best. This indicates that the bias correction methods based on point-wise or theoretical distribution are better than methods based on 100 quantile divisions for the correction of standard deviation. This may be because the 20 year time series is not enough to define 100 points in the empirical distribution.

[22] The mean-based bias correction methods (LS and LOCI) do not specifically consider the daily precipitation extremes. The heavy precipitation (95th percentile of wet-day precipitation in this study) is adjusted using the same correction factor as for light precipitation. Therefore, the heavy precipitation corrected by LS is similar to that without bias correction (Figure 5). Even though the LOCI method is also mean based, it improves the RCM-simulated heavy precipitation. This is because the LOCI method corrects the wet-day frequency, and thus, the corrected precipitation time series has less wet days than the LS corrected

time series, resulting in a larger value of the 95th percentile. In contrast, the other four bias correction methods specifically take into account the bias difference for different precipitation levels. All the methods are, therefore, able to improve the heavy precipitation simulation. The DT method is worse than the other three methods, due to the lack of correction in wet-day frequency. Similarly to the mean wet-day precipitation, even though the biases of heavy precipitation are RCM dependent, the DBC, QME, and QMG methods are able to reduce them to a similar level, indicating the robust performance of these three bias correction methods.

4.2. Hydrological Modeling

[23] The impacts of bias correction methods on hydrological modeling are evaluated by running the hydrological model HSAMI using the raw RCM-simulated and bias-corrected precipitation time series for all 10 river basins. The streamflows simulated using the raw RCM-simulated precipitation with the specific calibration of each hydrological model are also computed. To avoid any bias resulting from the hydrological modeling process, the reference streamflows are represented by the modeled streamflows using observed precipitation and temperature and not by observed discharge. Prior to comparing the performance of the bias correction methods in hydrological modeling, the results of the specific calibration and validation of HSAMI using RCM-simulated precipitation combined with observed Tmax and Tmin are presented for all 10 river basins.

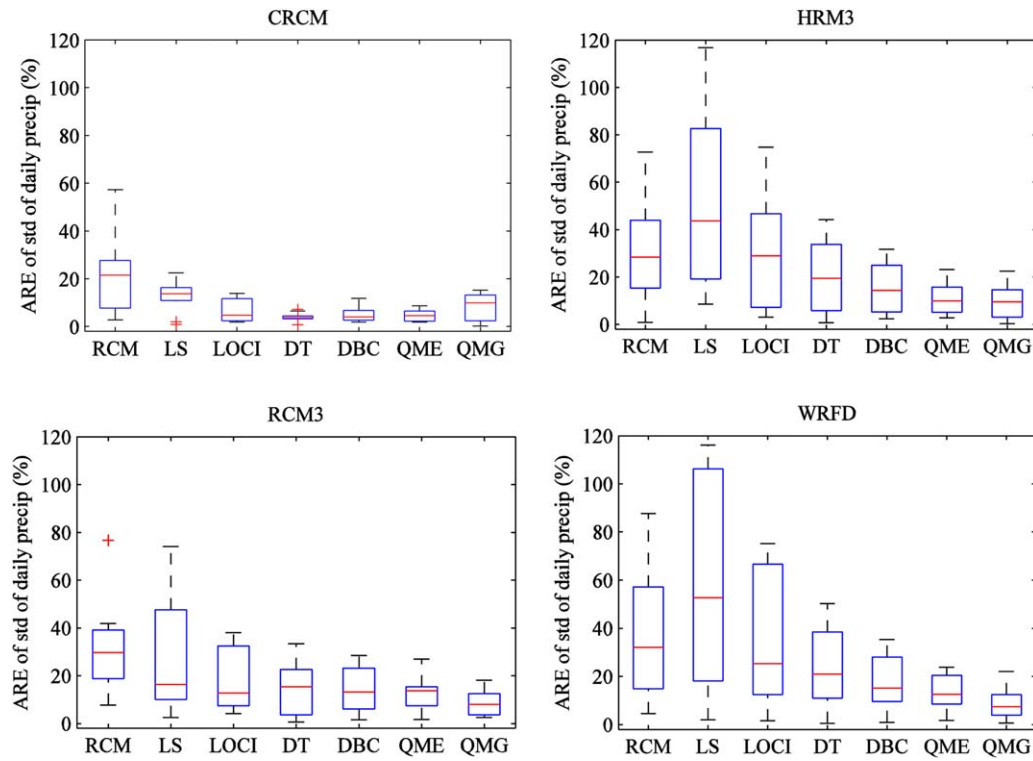


Figure 4. Boxplot of the ARE of standard deviation (std) of raw RCM-simulated (RCM) and six bias correction method-corrected daily precipitation (precip) values for all four RCMs. Each boxplot is constructed using 10 standard values from 10 river basins.

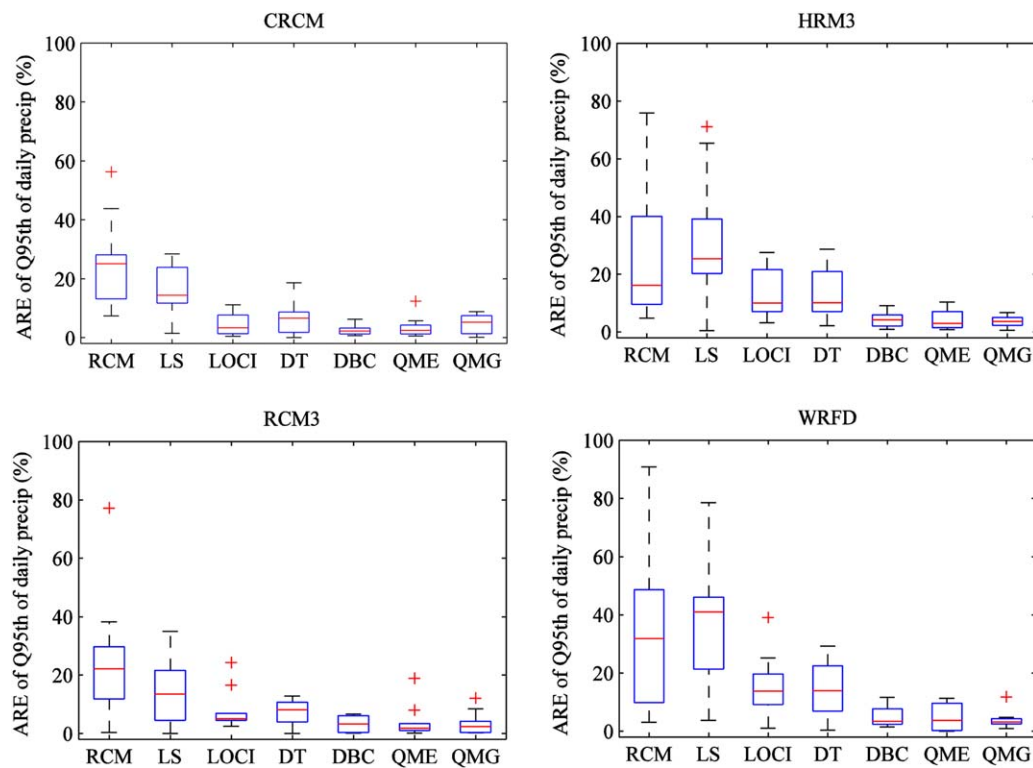


Figure 5. Boxplot of the ARE of the 95th percentile of raw RCM-simulated (RCM) and six bias correction method-corrected daily precipitation (precip) values for all four RCMs. Each boxplot is constructed using 10 values from 10 river basins.

Table 3. Nash-Sutcliffe Coefficient (NSE) of the Calibration (CAL) and Validation (VAL) of Hydrological Model HSAMI Using RCM-Simulated Precipitation Combining With Observed Temperature (Odd Year for Calibration and Even Year for Validation)^a

Model	Source	Nation	Carrot	Manic5	Umpqua	Yellowstone	Yampa	Grand	Wolf	Chickasawhay	Sacandaga	Mean
CRCM	CAL	0.84	<i>0.58</i>	0.86	0.75	0.92	0.85	0.27	<i>0.48</i>	0.39	<i>0.54</i>	0.65
	VAL	0.85	<i>0.37</i>	0.77	0.70	0.78	0.73	-0.11	<i>0.46</i>	0.33	<i>0.18</i>	0.51
HRM3	CAL	0.91	<i>0.73</i>	0.78	0.64	0.88	0.72	0.07	<i>0.28</i>	0.29	<i>0.39</i>	0.57
	VAL	0.89	<i>-0.95</i>	0.76	0.61	0.79	0.41	0.06	<i>0.30</i>	0.00	<i>0.34</i>	0.32
RCM3	CAL	0.81	<i>0.38</i>	0.78	0.55	0.89	0.80	0.09	<i>0.24</i>	-0.07	<i>0.51</i>	0.50
	VAL	0.81	<i>0.18</i>	0.76	0.58	0.75	0.68	0.02	<i>0.23</i>	-0.14	<i>0.31</i>	0.42
WRF3	CAL	0.79	<i>0.34</i>	0.69	0.61	0.87	0.77	0.04	<i>0.26</i>	0.22	<i>0.47</i>	0.51
	VAL	0.86	<i>0.06</i>	0.73	0.39	0.78	0.73	0.03	<i>0.05</i>	0.14	<i>0.35</i>	0.41
Mean		0.84	<i>0.21</i>	0.77	0.60	0.83	0.71	0.06	<i>0.29</i>	0.15	<i>0.39</i>	

^aDifferent typefaces on watershed name represent different performances of bias correction approaches for hydrological modeling: roman = good, italic = bad, and bold = very bad.

4.2.1. Specific Calibration and Validation of HSAMI

[24] According to the method presented in section 3.2, the hydrological model HSAMI was calibrated using raw RCM-simulated precipitation combined with observed Tmax and Tmin. Since the RCM-simulated precipitation only covers 20 years, 10 odd years were used for calibration and 10 even years for validation. The results are presented in Table 3. The specific calibration of HSAMI using raw RCM-simulated precipitation indicates that the biases are small enough to be partly accounted for by the hydrological model for five out of 10 basins. However, the obtained NSE values were lower than those of the standard calibration using observed data, indicating that the RCM-simulated precipitation is either more biased than the observed values or less coherent with the observed discharge. However, the NSE values are very low for the other five basins (Carrot, Grand, Wolf, Chickasawhay, and Sacandaga), and especially low for Grand and Chickasawhay, indicating that the RCM data are either very biased or not coherent with the observed discharge for these five basins (or a combination of both). The HSAMI calibrated using CRCM-simulated precipitation is consistently better than the other three RCMs, indicating the better performance of CRCM at representing the observed precipitation (occurrence and amount). This is likely because the CRCM includes spectral nudging to improve its performance. The performance of the hydrological model calibrated using raw RCM-simulated precipitation is considered to be “good” for five basins (Nation, Manic5, Umpqua, Yellowstone, and Yampa), “bad” for three basins (Carrot, Wolf, and Scandaga, denoted by italics in Tables 3–5 and in Figure 1), and “very bad” (Grand and Chickasawhay, denoted by bold in Tables 3–5 and in Figure 1) for two basins.

4.2.2. Performance of Bias Correction Methods in Hydrological Modeling

[25] Figure 6 presents the mean annual hydrographs using the CRCM outputs for all 10 river basins. The results for the other three RCMs are not shown. Overall, raw RCM-simulated precipitation is unable to reproduce the observed streamflow for most river basins. However, the results of snow-dominated basins are generally better than those of the no-snowfall basins, with the exception of Nation and Carrot, where peak discharges are considerably overestimated. In particular, RCM-simulated precipitation is not capable of representing the hydrological cycle for the Chickasawhay basin. Even for the snow-dominated basins, the simulated streamflow is very different when using the

RCM-simulated precipitation. For example, RCM-simulated precipitation reasonably reproduces the observed mean annual hydrograph for the Manic 5 basin, while it significantly overestimates the peak discharge for both the Nation and Carrot basins. With the specific calibration of a hydrological model, the simulated streamflow is much improved compared to the standard hydrological model calibration, indicating that the biases of RCM-simulated precipitation can be somewhat overcome by a hydrological model through the calibration process.

[26] All bias correction methods are able to improve the simulation of mean annual streamflows for all basins. The mean annual hydrographs simulated using corrected data match those simulated using observed data very well. From the mean annual hydrograph, it is difficult to see the differences between the bias correction methods in terms of hydrological modeling. These differences will be assessed using hydrological statistics and evaluation metrics in the following sections.

[27] Figure 7 presents the ARE of the mean daily discharge simulated using raw RCM-simulated and six bias-corrected daily precipitation time series for all four RCMs. AREs of the mean daily discharge simulated using raw RCM-simulated precipitation with the specific calibration of a hydrological model are also plotted for comparison. The mean daily discharge simulated using raw RCM-simulated precipitation is very biased, with the median of AREs ranging between 18.6% and 41.5% for all four RCMs. However, all the bias correction methods are able to improve the simulation of mean daily discharge for all four RCMs. The distribution-based methods (DT, DBC, QME, and QMG) are consistently better than the mean-based methods (LS and LOCI), with the exception of the LOCI method for CRCM. In particular, the mean daily discharges simulated using the LS and LOCI method’s corrected precipitation are even worse than those simulated using raw RCM-simulated precipitation with the specific calibration of HSAMI for three out of four RCMs. This is because the hydrological model was calibrated with different data sets: Observed precipitation for the former and RCM-simulated precipitation for the latter. For the two mean-based methods, the LOCI method with the correction in wet-day frequency is consistently better than the LS method. However, there is no obvious difference between the DT and the DBC method in producing the mean daily discharge, even though they differ in how they deal with the wet-day frequency. Moreover, the bias correction

Table 4. The Nash-Sutcliffe Coefficient (NSE) Between Discharges Simulated Using Observed (OBS) and Raw RCM-Simulated (RCM), and Six Bias Correction Methods' Corrected Precipitation Time Series for All Four RCMs and 10 River Basins^a

Model	Method	Nation	Carrot	Manic5	Umpqua	Yellowstone	Yampa	Grand	Wolf	Chickasawhay	Sacandaga
CRCM	OBS	0.91	0.72	0.86	0.89	0.70	0.73	0.78	0.74	0.85	0.81
	RCM	-0.62	0.09	0.82	0.32	0.82	0.81	-0.16	0.03	0.03	0.40
	RCMC	0.86	0.48	0.81	0.69	0.78	0.83	-0.02	0.49	0.34	0.33
	LS	0.90	0.51	0.83	0.67	0.95	0.88	-0.05	0.41	0.31	0.33
	LOCI	0.89	0.49	0.83	0.67	0.95	0.89	-0.16	0.31	0.17	0.28
	DT	0.89	0.48	0.83	0.63	0.94	0.87	-0.33	0.15	0.10	0.39
	DBC	0.90	0.45	0.82	0.63	0.94	0.88	-0.33	0.12	0.10	0.39
	QME	0.89	0.51	0.83	0.63	0.94	0.88	-0.22	0.25	0.21	0.40
HRM3	QMG	0.88	0.49	0.86	0.64	0.95	0.90	-0.33	0.13	0.08	0.28
	RCM	-2.38	-0.32	0.66	0.47	0.37	0.66	-0.65	-0.63	-0.18	-4.52
	RCMC	0.84	-0.56	0.77	0.60	0.75	0.54	-0.01	0.23	0.11	0.36
	LS	0.92	-2.62	0.63	-0.25	0.84	0.51	-2.05	-3.12	-0.94	0.37
	LOCI	0.92	-1.73	0.66	0.07	0.86	0.55	-1.80	-2.22	-0.88	0.31
	DT	0.92	-0.30	0.70	0.49	0.83	0.64	-1.10	-0.93	-0.62	0.21
	DBC	0.92	-0.50	0.68	0.49	0.83	0.64	-1.13	-1.00	-0.63	0.19
	QME	0.92	-0.49	0.68	0.50	0.88	0.55	-0.81	-0.59	-0.42	0.14
RCM3	QMG	0.93	0.01	0.75	0.52	0.86	0.64	-0.69	-0.45	-0.40	0.14
	RCM	-2.75	-8.71	0.25	-0.29	0.85	0.29	-0.79	-4.90	-0.91	0.27
	RCMC	0.77	0.23	0.81	0.56	0.75	0.74	0.04	0.14	-0.16	0.40
	LS	0.89	0.30	0.75	0.51	0.84	0.28	-1.52	-3.36	-0.72	0.39
	LOCI	0.88	0.22	0.73	0.51	0.88	0.76	-1.12	-2.75	-0.66	0.32
	DT	0.86	0.15	0.73	0.45	0.90	0.74	-0.70	-1.84	-0.17	0.28
	DBC	0.88	0.05	0.71	0.45	0.90	0.73	-0.71	-1.92	-0.17	0.26
	QME	0.88	0.15	0.73	0.35	0.89	0.79	-0.68	-1.91	-0.12	0.32
WRF3	QMG	0.88	0.19	0.75	0.43	0.91	0.83	-0.42	-1.19	-0.11	0.29
	RCM	-1.39	-0.74	0.58	-0.12	0.72	0.76	-0.33	-0.59	-0.18	0.32
	RCMC	0.79	-0.04	0.73	0.43	0.73	0.72	-0.02	0.12	0.16	0.45
	LS	0.86	-3.06	0.58	0.39	0.85	0.73	-4.79	-6.04	-2.85	0.19
	LOCI	0.86	-1.99	0.64	0.43	0.88	0.82	-3.78	-3.79	-2.26	0.22
	DT	0.87	-0.62	0.73	0.43	0.75	0.87	-2.12	-0.75	-0.51	0.28
	DBC	0.87	-0.87	0.73	0.43	0.74	0.86	-2.15	-0.79	-0.52	0.28
	QME	0.86	-0.91	0.68	0.36	0.84	0.84	-1.34	-0.97	-0.38	0.32
	QMG	0.81	-0.53	0.73	0.42	0.87	0.87	-0.99	-0.43	-0.30	0.33

^aThe NSE values between discharges simulated using observed and RCM-simulated precipitation time series' with the specific calibration of hydrological model (RCMC) are also presented. Different typefaces on watershed name represent different performances of bias correction approaches for hydrological modeling: Black = good, blue = bad, and red = very bad.

methods based on point-wise or theoretical distribution are similar to quantile-based methods based on 100 quantiles in the production of mean daily discharge. In the case of the HRM3, RCM3, and WRF3 climate models, it is interesting to note the presence of outliers. The outliers indicate that the bias correction methods did not perform well on at least one basin. The presence of these outliers will be discussed later.

[28] Similarly to Figure 7, Figure 8 presents the AREs of mean spring high flows (Q95) (mean value of 20 years of

spring high flow) for all four RCMs. The AREs are very large for mean spring high flow simulated using raw RCM precipitation, with medians of AREs ranging between 24.6% and 50.6%. All of the bias correction methods are able to improve the RCM-simulated precipitation in the representation of mean spring high flow, with AREs ranging between 4.3% and 14.6% across all six bias correction methods and four RCMs. The specific calibration of a hydrological model improves the RCM-simulated precipitation in the representation of mean spring high flow, which

Table 5. Percentile (%) of Correct Wet and Dry Days for RCM-Simulated Daily Precipitation With a Correction in Wet-Day Frequency for All 10 River Basins^a

Model	Source	Nation	Carrot	Manic5	Umpqua	Yellowstone	Yampa	Grand	Wolf	Chickasawhay	Sacandaga
Wet day frequency		52.2	34.1	63.8	47.6	62.1	56.5	37.0	39.5	44.3	48.6
CRCM	Correct wet day	67.9	53.9	75.5	81.2	76.4	73.1	60.9	61.6	63.9	64.7
	Correct dry day	65.0	76.1	56.7	82.9	61.3	64.9	77.0	74.8	71.2	66.7
HRM3	Correct wet day	67.8	44.5	67.7	72.1	70.9	63.1	49.6	48.4	53.4	53.8
	Correct dry day	64.8	71.3	43.0	74.7	52.3	52.0	70.4	66.2	62.9	56.4
RCM3	Correct wet day	65.6	49.1	69.4	77.2	72.8	66.3	52.7	50.8	56.4	56.7
	Correct dry day	62.5	73.7	46.0	79.3	55.4	60.6	73.5	67.8	65.4	59.1
WRF3	Correct wet day	64.9	47.1	69.4	75.9	72.2	67.5	49.7	50.6	56.7	57.8
	Correct dry day	61.8	72.7	46.9	78.1	54.3	57.6	70.8	68.3	66.5	60.0

^aThe time period covered is 1981–2000. Different typefaces on watershed name represent different performances of bias correction approaches for hydrological modeling: Black = good, blue = bad, and red = very bad.

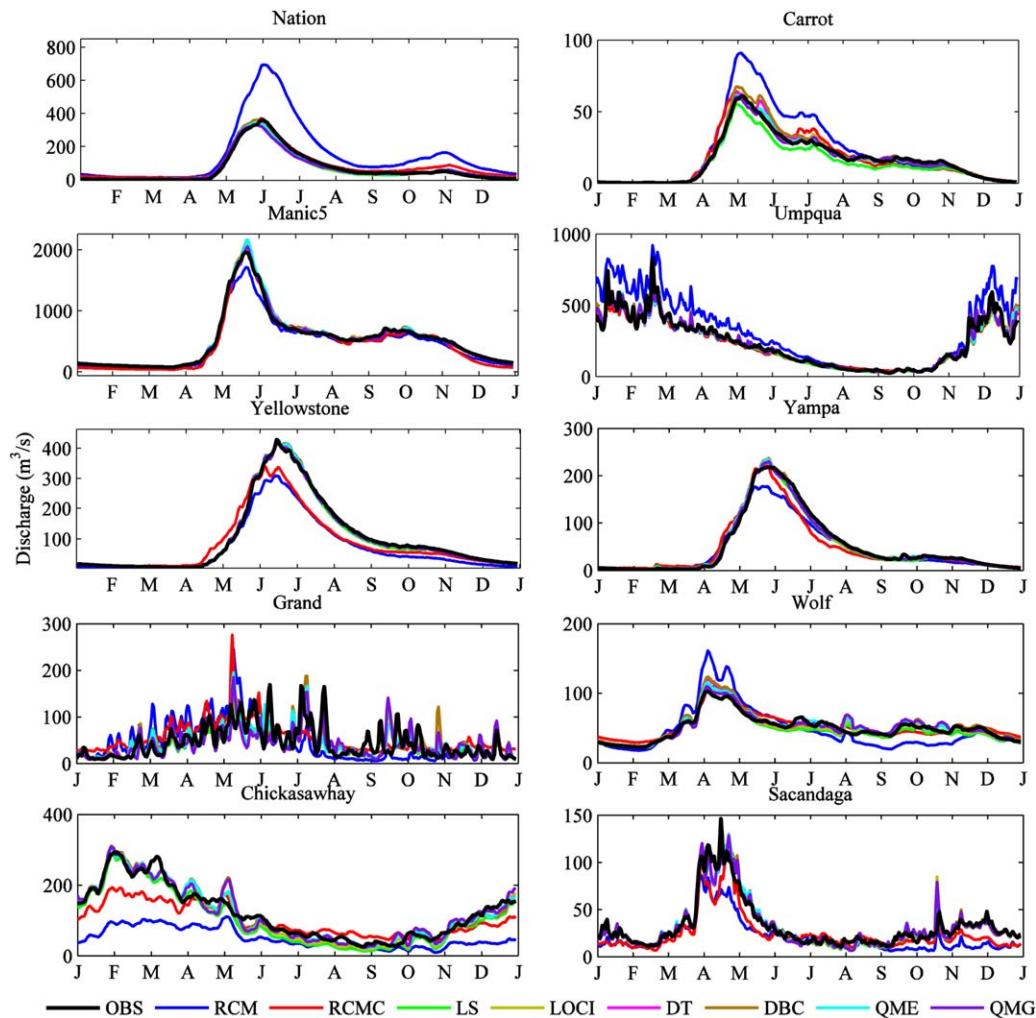


Figure 6. Mean annual hydrographs simulated using observed (OBS), raw RCM-simulated (RCM), and six bias correction methods' corrected precipitation time series from the CRCM simulation for all 10 river basins. The simulated hydrograph using RCM-simulated precipitation (RCMC) but with the specific hydrological model calibration is also plotted.

is comparable to that of most bias correction methods. Again, the four distribution-based methods are consistently better than the two mean-based methods, and the LOCI method with the correction in wet-day frequency is consistently better than the LS method. However, there is no obvious difference among the four distribution-based methods.

[29] Figure 9 presents the AREs of mean summer low flows (Q5) (mean value of 20 years of summer low flow). When using raw RCM-simulated precipitation, with either the standard or specific hydrological model calibration, the AREs of mean summer low flows are very large, with medians ranging between 32.5% and 61.7% for the standard hydrological model calibration and between 22.0% and 58.0% for the specific calibration. All the bias correction methods considerably improve the precipitation in the simulation of mean summer low flow, with medians of AREs ranging between 2.3% and 13.3% for all six bias correction methods and four RCMs. The river basin variability is very small for all the bias correction methods, indicating that the bias correction methods are able to improve the precipita-

tion in the representation of mean summer low flow for all river basins. The distribution-based methods are slightly better than the mean-based methods for three out of four RCMs. Furthermore, there is no distinct difference within each type of bias correction method.

[30] The use of an RCM driven by reanalysis data makes it possible to evaluate the coherence between observed and simulated streamflow time series' at the daily scale. They can be evaluated using metrics such as the NSE, RMSE, and TRMSE. Since similar results were obtained for all metrics, only the results from using of the NSE are presented in Table 4 for illustration. The second row in Table 4 presents the NSE values of the HSAMI validation for the 1981–2000 period.

[31] The NSE values between discharges simulated using raw RCM-simulated (labeled RCM) precipitation are reasonably accurate for five out of 10 basins. This indicates that the bias of RCM-simulated precipitation is small or the hydrological model is not sensitive to the bias for these river basins. The direct use of RCM data with the specific calibration of hydrological model (labeled RCMC) somewhat

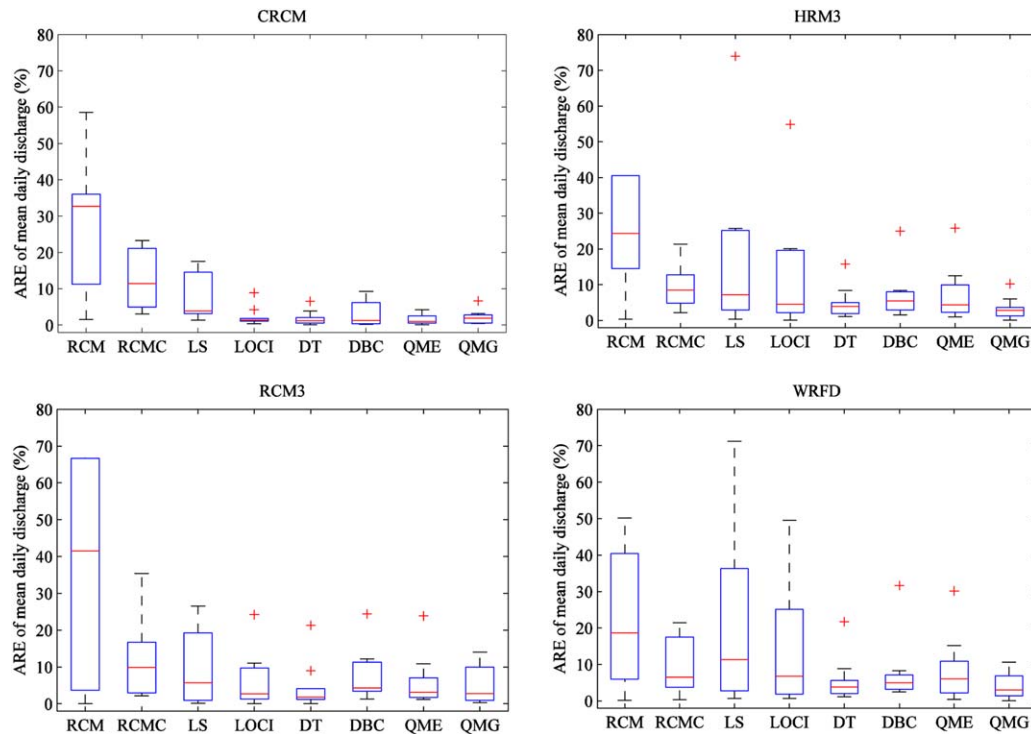


Figure 7. Boxplots of the ARE of the mean daily discharge simulated using raw RCM-simulated (RCM) and six bias correction methods' corrected daily precipitation time series for all four RCMs. The ARE of the mean daily discharge simulated using a raw RCM-simulated precipitation time series with the specific calibration of hydrological model RCMC is also plotted. Each boxplot is constructed from the values of 10 river basins. For plotting purposes, one or two outliers of the boxplot constructed by the ARE of mean daily discharge simulated using raw RCM-simulated precipitation have been omitted.

improves the simulation of streamflow for all river basins and RCMs. However, similarly to the calibration and validation of hydrological model presented in section 4.2.1, results are river basin dependent. As discussed earlier, the 10 basins can be classified into three categories: “good” for five basins (Nation, Manic5, Umpqua, Yellowstone, and Yampa), “bad” for three basins (Carrot, Wolf, and Scandaga), and “very bad” for two basins (Grand and Chickasawhay). The “bad” and “very bad” results indicate that the RCM-simulated precipitation is not coherent with the observed streamflows and the specific calibration of the hydrology model is unable to overcome the biases.

[32] Bias correction methods somewhat improve the RCM-simulated precipitation in the representation of streamflows. Compared to the streamflows simulated using raw RCM-simulated precipitation with the standard calibration of hydrological model, the NSE values are consistently improved when using bias correction methods, even though the performance is basin dependent. For example, the NSE values are less than zero for the Nation basin when using RCM-simulated precipitation, while they are all greater than 0.8 when using corrected precipitation for all the RCMs. It should be noted that while bias correction methods are able to improve the precipitation quantity, they are unable to improve the coherence between modeled and observed time series. Thus, although the results are slightly improved for the three “bad” and two “very bad” basins, the NSE values are still very low, even less than zero for the two “very bad” basins. Similarly, the distribution-

based methods (DT, DBC, QME, and QMG) are better than the mean-based methods (LS and LOCI). The methods based on point-wise or theoretical distribution (QME and QMG) are slightly better than methods based on 100 quantile divisions (DT and DBC). The mean NSE values are -0.367 , -0.154 , 0.178 , 0.153 , 0.209 , and 0.289 for the LS, LOCI, DT, DBC, QME, and QMG methods, respectively, across all 10 river basins and four RCMs. Moreover, the results from the CRCM climate model are consistently better than those from the other three RCMs, with the exception of the Nation basin where HRM3 has similar results to those of the CRCM. The bias correction methods are robust for the “good” basin, but not robust for “very bad” basins.

5. Discussions

[33] In the last decade, several bias correction methods have been proposed to downscale climate model outputs (usually precipitation and temperature), and ultimately, for use in assessing climate change impacts. These methods range from simple LS approaches to sophisticated distribution mapping approaches. As such, a comparative performance evaluation is needed to properly conduct impact studies. This work evaluates and compares the performance of six bias correction methods over 10 North American watersheds. The performance is based on the ability to reproduce precipitation as well as river streamflows with the use of a hydrology model. Ten river basins across Canada and the US are selected to represent various climate

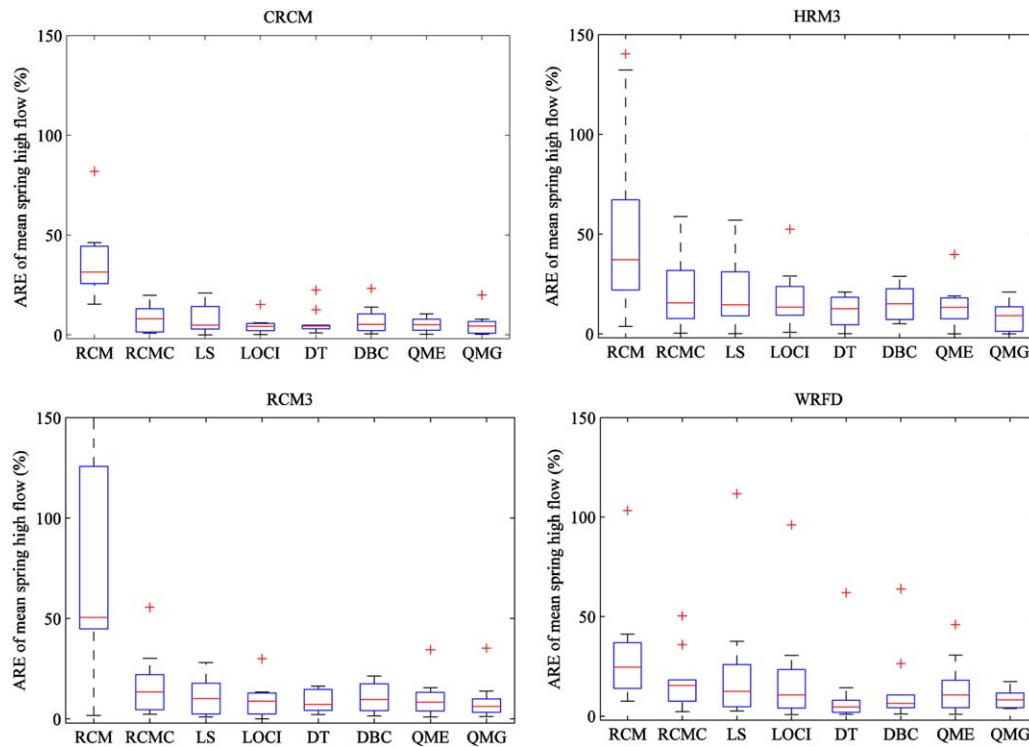


Figure 8. Boxplots of the ARE of mean spring daily high flow (95th percentile) simulated using raw RCM-simulated (RCM) and six bias correction methods' corrected daily precipitation time series for all four RCMs. The ARE of the mean spring daily high flow simulated using RCM-simulated precipitation time series with the specific calibration of a hydrological model (RCMC) is also plotted. Each boxplot is constructed using values from 10 river basins. For plotting simplicity, one part of the Whisker of a boxplot constructed by the ARE of mean spring high flow simulated using raw RCM3-simulated precipitation have been omitted.

and topographic conditions. The direct use of RCM data with standard and specific calibration of the hydrological model is also presented for comparison. The six bias correction methods can be classified into mean-based (two methods) and distribution-based (four methods) approaches. They can also be classified into methods with and without correction in the wet-day frequency. Four RCMs driven by NCEP reanalysis data are used to analyze the sensitivity of bias correction methods to climate models. Using RCMs driven by reanalysis data enables the coherence between the corrected and observed time series to be analyzed.

[34] As discussed in several studies [Jacob *et al.*, 2007; Kjellström *et al.*, 2005, 2011; Rummukainen *et al.*, 2001; Teutschbein and Seibert, 2010, 2012], RCM-simulated daily precipitation is usually biased, sometimes severely. In this study, the biases displayed by the four RCMs are different, indicating the model biases are mostly derived from systematic RCM error, and not by biases in the reanalysis data.

[35] All the bias correction methods evaluated are able to somewhat improve the raw RCM-simulated precipitation. However, the performance for precipitation simulation depends on the choice of a bias correction method. LS is the simplest bias correction method, which adjusts monthly mean precipitation based on the difference between observed and RCM-simulated mean precipitation for a specific month.

As presented in Table 2, the disadvantage of LS is that all precipitation events in a month are adjusted with the same correction factor and there is no correction in the wet-day frequency (RCMs usually simulate too many wet days and the occurrence of light precipitation—drizzle effect). Therefore, LS has limited improvement ability for precipitation statistics such as mean, standard deviation, and heavy precipitation. In fact, the standard deviation is worsened for three out of four RCMs. This indicates that bias correction that acts on the monthly scale, such as LS, has a limited effect on properties at the daily scale. Haerter *et al.* [2011] pointed out that “the improvements to the statistical properties of the data are limited to the specific timescale of the fluctuations that are considered.” LOCI is an extended version of the LS method, as it takes into account the overestimation of precipitation occurrence simulated by RCMs. This is important, especially when a relatively large precipitation threshold is used for the observed precipitation, such as the 0.5 mm used in this study. Thus, LOCI considerably improves the RCM-simulated precipitation, and in this study, especially the mean and heavy precipitation, which is comparable to the distribution-based correction method. However, the influence on standard deviation is still limited. For the distribution-based methods, different correction factors are applied to precipitation at different levels for a specific month. However, the DT does not consider the overestimation of precipitation occurrence simulated by

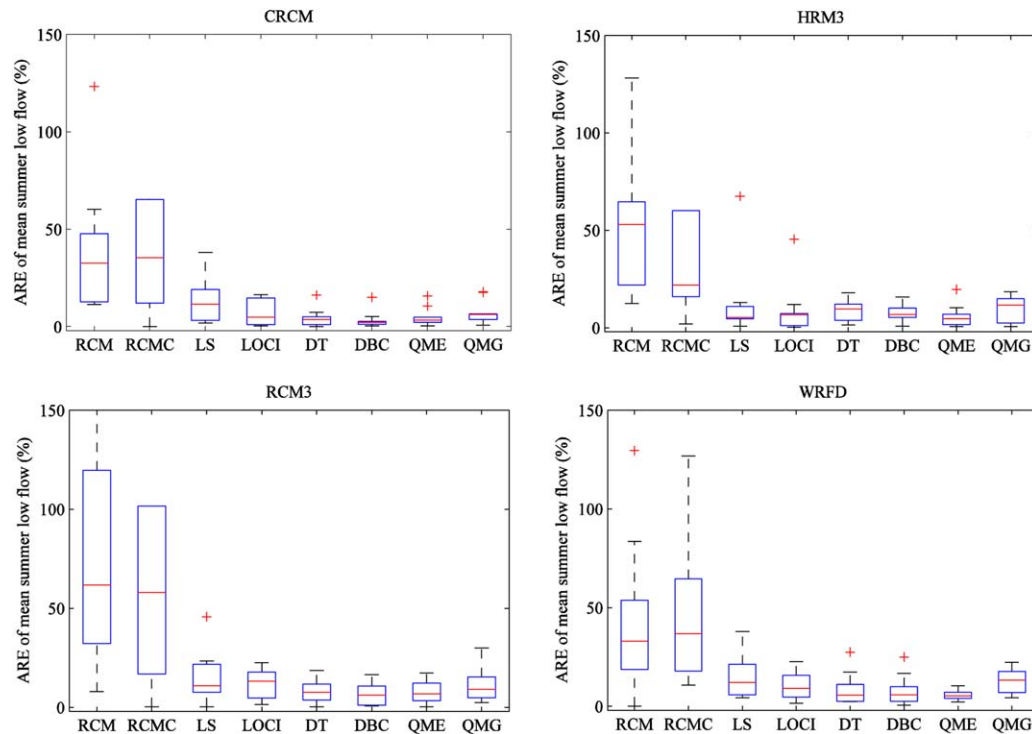


Figure 9. Boxplots of the ARE of mean summer daily low flow (5th percentile) simulated using raw RCM-simulated (RCM) and six bias correction methods' corrected daily precipitation time series for all four RCMs. The ARE of the mean summer daily low flow simulated using an RCM-simulated precipitation time series with the specific calibration of hydrological model (RCMC) is also plotted. Each boxplot is constructed using the values of 10 river basins. For plotting simplicity, one or two outliers or part of the Whiskers of boxplots constructed by the ARE of mean summer low flow simulated using raw RCM-simulated precipitation (or using raw-RCM simulated precipitation but with specific calibration of hydrological model) have been omitted.

RCMs, while, like the LOCI method, the other three methods correct the wet-day frequency by setting the modeled precipitation values below a certain threshold to zero. The DT method without the correction in wet-day frequency results in a large bias in the mean daily precipitation. Both the DT and the DBC methods correct daily precipitation based on 100 quantile divisions, and the correction factors among these 100 quantiles are interpolated. Lafon *et al.* [2012] showed that empirical distribution mapping with 100 quantile divisions was more accurate than methods with 25, 50, or 75 divisions. Therefore, both these methods are consistently better than the LS and LOCI methods that correct all the wet-day precipitation in a month based on only one correction factor. Both the QME and QMG methods take into account different precipitation intensities individually, based on point-wise empirical and gamma distributions, respectively. In this study, these two perform the best. In particular, they take into account the possible changes in extreme events. One challenge of using the QME method is to deal with the “new extremes” (modeled values beyond the observed range). The commonly used constant extrapolation using only the single highest quantile for extremes may be invalid for the extreme tail of the precipitation distribution [Maraun *et al.*, 2010], especially when correction factors for the calibration period are unstable for extreme precipitation [Chen *et al.*, 2013]. Thus, similarly to the study of Chen *et al.* [2013], this study uses the mean of the 10

highest quantiles from the control simulation instead of the single highest quantile. This procedure may result in a slightly smoothed change in extreme events. Fitting theoretical distribution (such as gamma distribution) to the empirical distribution function of the precipitation resolved the “new extremes” problem. The prerequisite of using theoretical distribution implies that both observed and modeled data should follow the proposed distribution. The gamma distribution is the one most widely used to fit the daily precipitation. However, it is well known that the gamma distribution is usually inadequate to represent the extreme tail of daily precipitation's distribution. A possible solution could be to use a mixed distribution, such as the mixture of gamma and Pareto distribution used by Vrac and Naveau [2007].

[36] All bias correction approaches used in this study only focus on the correction of mean precipitation at different scales. Other important properties such as interannual variability were not specifically taken into account. It has been shown that climate models may not perfectly represent the low frequency of climate variability. For example, Johnson *et al.* [2011] assessed the skill of 23 GCMs in simulating persistence across multiple time scales and found that only two GCMs exhibited an appropriate representation of the global low-frequency variability for individual variables, when considering the global sea surface temperature, precipitation and surface pressure anomaly fields. Thus, it may ultimately be necessary to include the

correction of low-frequency variability in a bias correction approach. *Johnson and Sharma* [2009, 2011, 2012] proposed and tested a nested method for correcting precipitation variability at multiple scales. The results showed that the nested bias correction approach was successful in correcting the representation of interannual variability. Future precipitation projections were strongly influenced by the bias correction methods, especially when taking into account the interannual variability. However, the relatively short-time period of RCM-simulated precipitation and odd-year and even-year separation in this study restrict the application of the nested bias correction approach. For other hydrological impact studies such as drought studies, it may be important to represent the low-frequency variability, since wetter and dryer years are not random, but rather come in series.

[37] When the corrected precipitation is transferred to the river basin hydrology, considerable improvements are observed compared to the use of raw RCM-simulated precipitation. Again, the LOCI method with the correction of wet-day frequency is consistently better than the LS method for all hydrological statistics. Furthermore, all distribution-based methods are consistently better than the mean-based methods. All the distribution-based methods perform similarly well, even though the corrected precipitation series are slightly different. For example, the DT method without correction of wet-day frequency displays relatively lower performance in the mean daily precipitation than the other three distribution-based methods (Figure 3). At the same time, all four methods perform equally well for hydrological modeling, indicating that the drizzle effect may be not significant when quantifying the hydrological impacts. The same conclusion was also obtained from the study of *Chen et al.* [2013].

[38] Compared to the direct use of RCM data with the standard calibration of hydrological model, the hydrological model specifically calibrated using raw RCM-simulated data somewhat improves the simulation of streamflow. However, the performance is almost always worse than with bias-corrected precipitation, indicating that RCM data are more biased than the observed values. This is especially true in cases where the hydrological model is unable to be properly calibrated using raw RCM-simulated data. Therefore, the bias correction of RCM outputs is still necessary for hydrological impact studies at the current stage. In general, the results clearly indicate that distribution-based methods perform best.

[39] On the other hand, when looking at the coherence between the daily discharges simulated by observed and by corrected precipitation time series' (Table 4), it appears that bias correction does not perform equally well on all the watersheds. On some of the studied watersheds, the tested methods are unable to overcome the biases present in the RCM data. As mentioned earlier, the NSE values (similar results were also observed for RMSE and TRMSE) between daily discharges simulated using observed and corrected precipitation time series are "good" for five basins (Nation, Manic5, Umpqua, Yellowstone, and Yampa), "bad" for three (Carrot, Wolf, and Sacandaga) and "very bad" for the remaining two basins (Grand and Chickasawhay). This conclusion is consistent for all four RCMs, indicating coherence problems between observed,

modeled, and corrected precipitation time series. The lack of coherence may result from two sources: (1) the biases are not constant between the calibration and validation periods and/or (2) the RCM-simulated sequences of wet (and especially for high precipitation) and dry days driven by reanalysis data mismatch those of observed precipitation time series. To confirm these statements, the wet-day frequencies of RCM-simulated precipitation are corrected by setting the modeled precipitation values that fall below a certain threshold to zero for the whole 20 year (1981–2000) time series, while the wet-day precipitation amount is not adjusted. The bias of modeled precipitation $((P_{\text{mod}} - P_{\text{obs}})/P_{\text{obs}} \times 100)$ is calculated for both the odd and the even years (10 years for each). The successful rate of wet and dry days (modeled versus observed precipitation) was also calculated for the entire 20 year RCM time series. Since similar results are obtained for each of the river basins, only the biases for Chickasawhay (a "very bad" basin) and Nation (a "good" basin) are presented in Figure 10 for illustration. For the Nation basin, the biases between 10 odd years and 10 even years are not constant in the winter, but they are constant in the other seasons for all RCMs. However, the same trend is not observed for the Chickasawhay basin. The differences in biases between odd and even years are relatively larger and do not present a seasonal trend. As mentioned earlier, the hydrological simulation of these two river basins is very different. Nation is a snow-dominated basin with a few months of accumulating snowfall in the winter. Thus, the hydrology is not sensitive to the precipitation time series. There is no snowfall in the Chickasawhay basin, and so the precipitation directly contributes to the streamflows. Thus, a small bias in precipitation may result in a large deviation in the overall hydrology. It should be noted that the variability in bias may be partly due to the short-time period (10 years) used to calibrate the bias correction method in this study. This factor may be a limitation of this study. However, the division into odd and even years should remove any interannual differences.

[40] In fact, all RCMs fail to adequately simulate the wet and dry states for every river basin (Table 5), although there is some regional variability. Even though the wet-day frequency is the same between observed and RCM-simulated precipitation, the ratio of correct wet and dry days is less than 85% for all river basins. It should be noted that a perfectly random scheme would result in a 50% successful rate. The RCMs driven by reanalysis data are only slightly better than a random draw. The CRCM results are slightly better than the others, due to the implementation of spectral nudging in its computational scheme. The successful rate of wet and dry days is similar for all river basins except for the Umpqua. An obvious question: Why is the simulated streamflow "good" for five basins, and "bad" and "very bad" for the other five basins? Nation, Manic5, Yellowstone, and Yampa are snow-dominated basins; the snowfall is accumulated for a few months in the winter. Thus, the overall hydrology is less sensitive to the time sequences of wet and dry days. However, there is no snowfall in the winter for the Grand and Chickasawhay basins. The rainfall affects the overall hydrology with brief delays. Thus, successful simulations of wet and dry days are more important for such basins. Even though there is snowfall in the winter

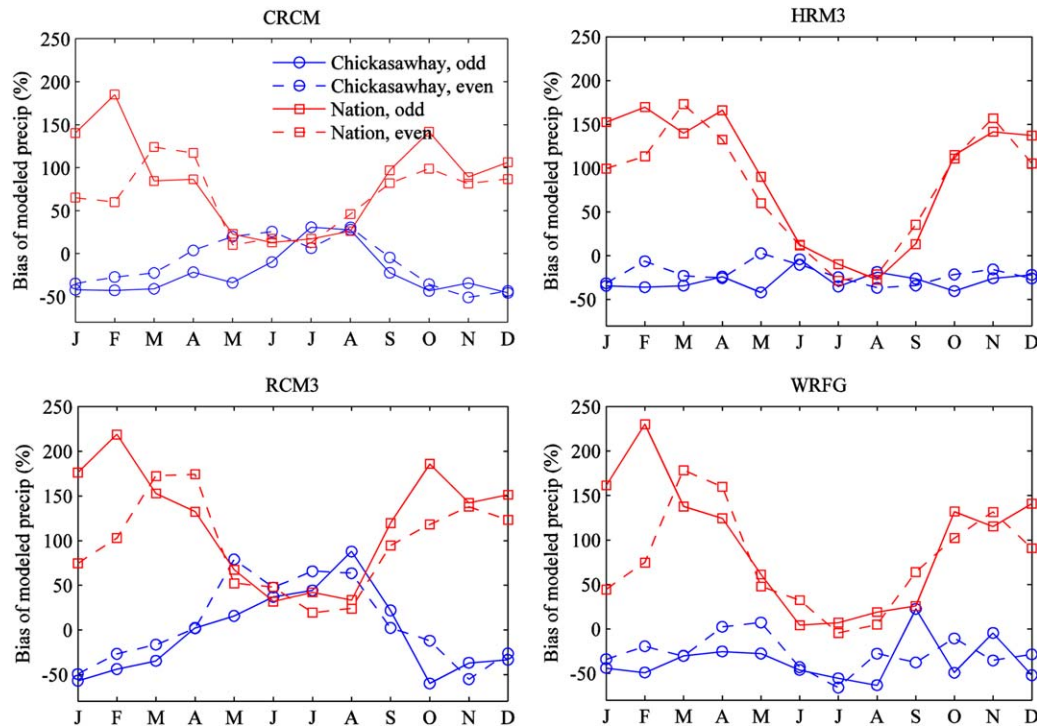


Figure 10. Bias of RCM-simulated precipitation (precip) compared to bias observations for Chickasawhay and Nation river basins, 20 year precipitation series is divided into odd and even years.

for the Wolf and Sacandaga basins, the snowfall is much less than in the five “good” basins, due to their southern locations. Additionally, these two basins, as well as the Grand and the Chickasawhay, are located in the central and southern US, close to the center of the North America computational domain (Figure 1) used in the NARCCAP inter-comparison study. The climate forcing from the boundary of the NCEP reanalysis data is stronger in the west and becomes weak to the center and east. Thus, the performance of RCMs’ simulation is also better in the west than in the center and east. A good example is the Umpqua basin, which is the only basin without snowfall in the winter that has “good” results from its hydrological simulation using corrected precipitation. The other factor may be that the Umpqua is located in a mountainous region. The horizontal flow from the NCEP forcing is forced to go from a low elevation to a higher elevation resulting in orographic precipitation. Orographic precipitation is easier to model than precipitation in flatter regions. Carrot is a snow-dominated Nordic basin with the driest conditions of all 10 basins. Carrot’s fewer wet days (the wet-day frequency is less than 40%) makes it more sensitive to the input data. Essentially, further away from the boundaries, and in the absence of significant topography, the climate models operate with maximum degrees of freedom. While continental circulation is well modeled, local storm paths can easily occur several grid points north and south and completely miss a watershed. This explains why the mean hydrology (such as the mean hydrographs in Figure 2) can be reproduced quite well, while the model fails at the daily scale. In such cases, no bias correction method can work. For bias correction methods to work, the biases must be constant over time and a reliable temporal structure of precipitation occurrence

must exist. Having a bias that stays constant is the most common assumption made when using bias correction methods (and most other statistical downscaling approaches). However, the temporal structure of daily precipitation occurrence significantly affects the performance of bias correction and, to the authors’ knowledge, this is the first study to document it. Apparently, bias correction methods will not work everywhere, for example, in the Central and Southern US in this study. In other words, these findings indicate that bias correction approaches are sounder for certain basins than others. For example, it is easy to take RCM outputs from South Africa and apply a bias correction scheme so that they adequately represent the hydroclimatology of a watershed in the southern United States. But this would be nonsense for climate change study. While it may be tempting to extrapolate the results of this paper for climate change studies or to the direct bias correction of GCM data, it would probably be a bit reckless to do so. In this paper, boundary conditions to the RCM were strictly from reanalysis. When driven by a GCM, RCM biases are superimposed over the GCM biases introduced at the boundary conditions. This is a complex problem that has received little attention in the literature. Testing bias correction schemes over the recent past using reanalysis-driven RCMs is one way to assess the robustness of a RCM over a given watershed. Previous studies such as those of *Teutschbein and Seibert* [2012] using RCMs driven by GCMs or directly using GCMs only served to hide the biases in the temporal structure of modeled precipitation. Therefore, for regions where bias correction methods perform poorly, the use of change factor approaches (also called delta change approaches) that apply the climate change signal to the observed time series may be a better choice.

[41] To complicate things a bit more, the conclusion that bias correction methods do not work well in the South-Central parts of the United States is limited to this study. The use of a different climate model or a different computational grid may significantly improve the performance of bias correction methods due to the better representation of the temporal structure of daily precipitation they may provide. This study clearly outlines the importance of proper validation of bias correction methods prior to any climate change impact study.

6. Conclusions

[42] This work evaluated the performance of six bias correction methods in reproducing precipitation and streamflows over 10 North American river basins. The six methods range from simple LS to more complex distribution mapping techniques. The performance of these bias correction methods was assessed via their correction of four RCMs' simulated precipitation driven by NCEP reanalysis data. The following conclusions can be drawn:

[43] RCM-simulated daily precipitation is always biased, sometimes significantly, which precludes its direct use with hydrological models. The direct use of RCM data with a specific calibration of the hydrological model somewhat improves the hydrological simulation, indicating that part of the bias can be overcome by the hydrological model.

[44] All bias correction methods are able to somewhat improve the RCM-simulated precipitation. Performance depends on the choice of a correction method. The correction in wet-day frequency is important for climate projection, while much less significant for hydrological simulations. Moreover, the distribution-based methods are consistently better than the mean-based methods for both climate projections and hydrological simulations.

[45] Bias correction methods will be invalid if the temporal structure of precipitation is not well-reproduced by the climate models, especially at the daily scale. This is more likely to happen when less forcing is exerted on the climate model, in flatter regions or away from the model computational boundaries. This problem should be less severe in snow-dominated regions because the winter hydrology is less sensitive to the precipitation occurrence and mostly conditioned on temperature, which is much better simulated by climate models than precipitation.

[46] Bias correction methods should be validated over the "recent past" prior to any climate change impact study. If such a validation is not possible, impact studies should rely on simpler methods that correct the observed time series, such as the delta-change method.

[47] Finally, this work further emphasizes the importance of using several climate models and bias correction methods to include the overall uncertainty for hydrological impact studies. In particular, for climate change impact studies, the use of only one model or method could give misleading results.

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